Ananthu S

linkedin.com/in/linkedin.com/in/ananthu-s-103219212
+918590335316
ananthusuresh9112@gmail.com
Kerala, India

As a self-motivated and responsible person, I love solving problems, taking up challenges and exploring new technologies. At the same time I look forward to completing problematic and growth oriented tasks that will prove my abilities in this field.

Competencies

Programming Skills	Python, Numpy, Pandas, Scikit-learn, Tensorflow, Matplotlib, Seaborn, OpenCV, C/C++, HTML
Database	Mongodb, MySQL, Redis, Cassanndra, Neo4j
Frameworks	LangChain, HuggingFace Transformers
Large Language Models(LLMs)	Gemini, OpenAI
Development Tools	Pycharm, Jupyter Notebook, Google Colab, Idle
Operating Systems	Windows

Certifications

2024 **AI and Applied Data Science with Python,** Data Science Academy

Python Programming | Machine Learning | Deep Learning | Data Analysis | Predictive Analytics | Time Series Forecasting | Text Mining

Algorithms : SVM, Neural Network, Decision Trees, Random Forest, Linear Regression, Logistic Regression etc.

Tools : Python (Numpy, Pandas, Scikit-Learn, Tensorflow, Keras, PyTorch, NLP, OpenCV)

Experience

September 2024 - Present Data Science Intern - CURVELOGICS ADVANCED TECHNOLOGY SOLUTIONS

Completed

INTELLIGENT CRIME DETECTION AND INVESTIGATION SYSTEM WITH REAL-TIME AI ANALYSIS, Telegram Bot API, OpenAI, DeepFace, LangChain, ChromaDB, OpenCV

- Developed an AI-powered surveillance application for automated crime investigation, reducing suspect identification time by 70 percent using real-time facial recognition with 95 percent accuracy on a 10,000+ record MySQLcriminal database.
- Integrated a reactive AIagent to autonomously analyze evidence, trigger facial recognition, and query databases for matches.
- Implemented RAG-based search with ChromaDB and MySQL for instant criminal profile retrieval and AI-driven suspect profiling using OpenAI and Gemini to predict movements.
- Enabled real-time video analysis with OpenCV, processing CCTV footage in under 3 seconds per frame.
- Deployed an automated alert system, notifying 50+ stations via Telegram.
- Integrated Google Drive API for secure crime data storage, enhancing investigation workflows.

Python | Telegram Bot API | Google Drive API | DeepFace | OpenAI | LangChain | ChromaDB | MySQL | OpenCV

Completed

AI POWERED EMAIL ANALYSIS SYSTEM, Generative AI, OpenAI, LangChain, ChromaDB

- Developed an advanced email analysis application using Python, integrating LangChain and OpenAI technologies.
- Implemented Retrieval-Augmented Generation (RAG) techniques to enhance the accuracy and efficiency of email content analysis.
- Employed vectorization methods to convert emails into embeddings, facilitating rapid and precise semantic searches.
- Integrated ChromaDB for storing and querying vectorized email data, ensuring swift and scalable retrieval of critical information.

Generative AI | OpenAI | LangChain | ChromaDB

Completed

FOOTBALL PLAYER DETECTION AND MASKING, YOLO, Autoencoders, Attention, U-Net, ViT

- Created bounding boxes for players on a football field using YOLO for object detection to focus on player regions.
- Developed a mask generation model with Autoencoders, refining results using an attention mechanism for more accurate output.
- Incorporated U-Net architecture to enhance the mask quality by improving segmentation accuracy.
- Designed and trained a Vision Transformer (ViT)-based model for the most precise image masking results.

Python | YOLO | Autoencoders | U-Net | ViT

Completed

NFL FORMATION DETECTION, YOLO, OpenCV, Tensorflow, Roboflow

- Annotated over 5,000 images of NFL plays using Roboflow to create a custom dataset for training a YOLO-based model.
- Used YOLO for detecting offensive and defensive formations from images, with pre-processing and augmentation via OpenCV.
- Trained the model using TensorFlow to accurately predict NFL formations from input images.

Python | Tensorflow | YOLO | OpenCV | Roboflow

Completed

FACE RECOGNITION SYSTEM, DEEPFACE , Computer Vision, Python

- Developed a face recognition system using DeepFace to identify and verify faces from images sourced from a Kaggle dataset.
- Utilized pre-trained models for facial feature extraction and comparison, achieving high accuracy in recognition.
- Employed Python for data processing and analysis, ensuring efficient handling and evaluation of face recognition results.

Python | DeepFace | OpenCV | Kaggle

Completed

AUTOMATED JOB SEARCH , WEB SCRAPING WITH SCRAPY AND SELENIUM , Data Analysis, Python

- Developed a web scraper using Scrapy and Selenium to extract data from dynamic websites with JavaScript-rendered content.
- Utilized Pandas and NumPy for data manipulation, exporting results to CSV for analysis.
- Integrated Flask for building a simple API to serve scraped data.

Scrapy | Selenium | Python | Flask

Qualifications

2023 **BTech in Computer Science and Engineering**, APJ Abdul Kalam Technological College
2019 **Higher Secondary**, MGM Model School, Varkala
2017 **CBSE Class 10th**, MGM Model School, Varkala

Languages

English ●●●●○
Hindi ●●●●○
Malayalam ●●●●●

References

Brijesh Madhavan
CEO of Curvelogics Advanced Technology Solution,
@ brijesh@curvelogics.com
+919880103691